



Aula 5

Simplificação de Diagramas de Blocos

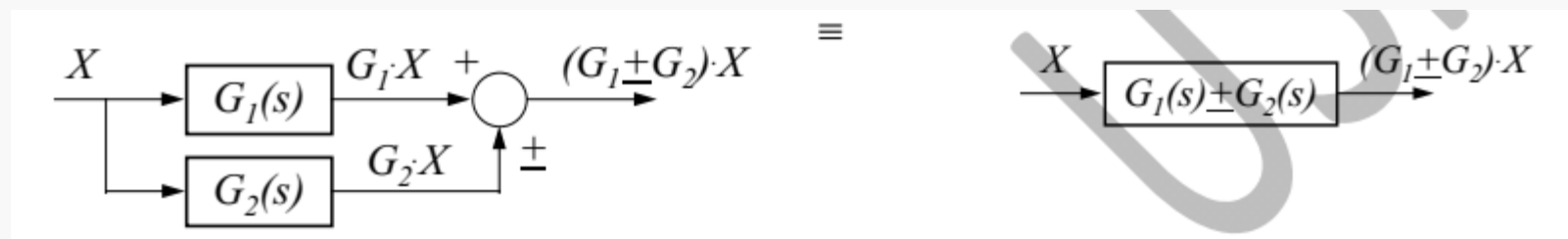
Prof. Ronaldo Guisso

Simplificação de Diagrama de Blocos

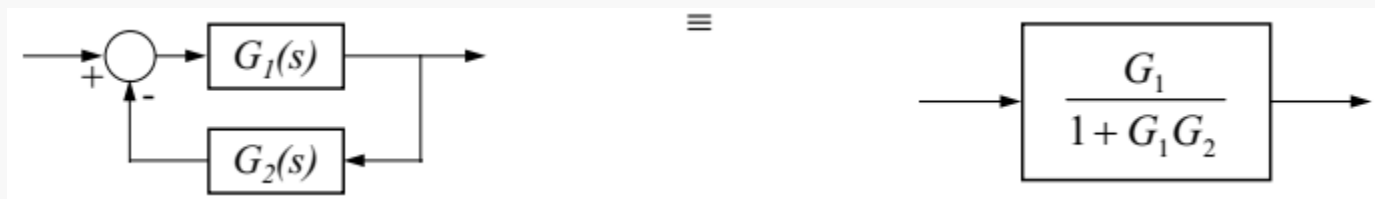
- Série ou Cascata:



- Paralelo:



- Realimentação (Feedback):



Simplificação de Diagrama de Blocos

- Movendo um somador para antes do bloco:



- Movendo um somador para depois do bloco:



Simplificação de Diagrama de Blocos

- Movendo um ponto de derivação para antes do bloco:



- Movendo um ponto de derivação para depois do bloco:

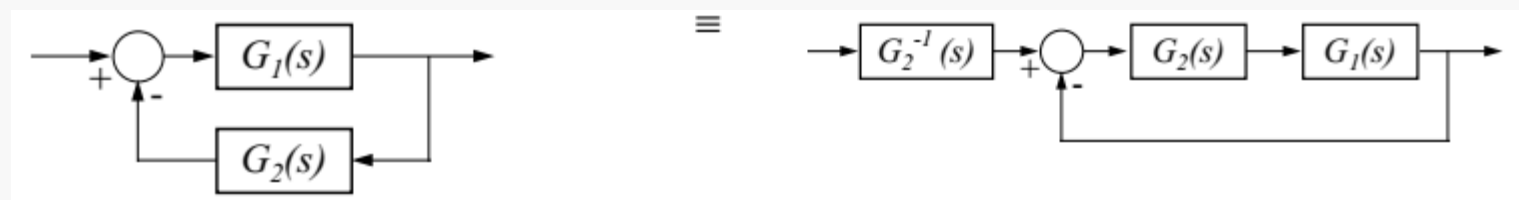


Simplificação de Diagrama de Blocos

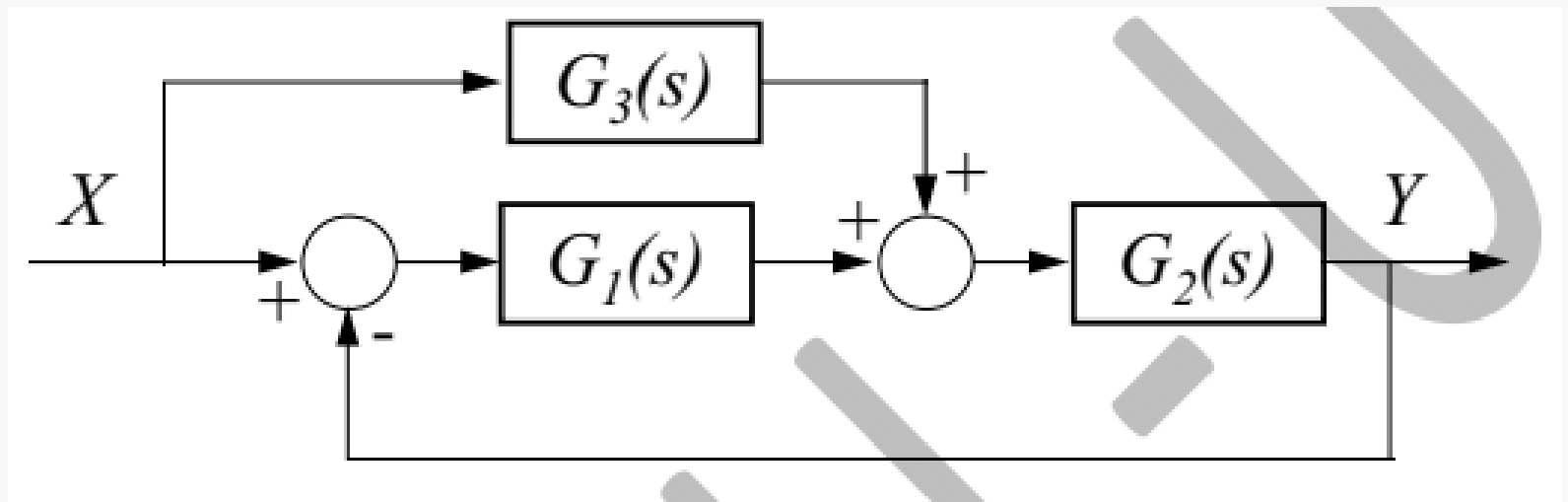
- Rearranjando somadores:



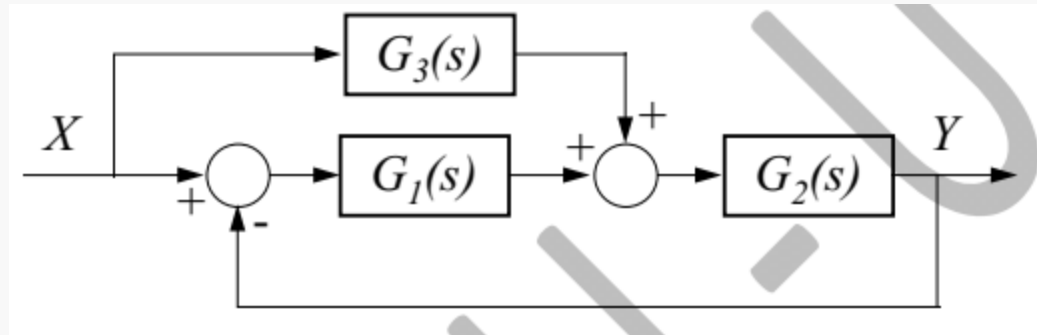
- Movendo o bloco de realimentação para antes do somador:



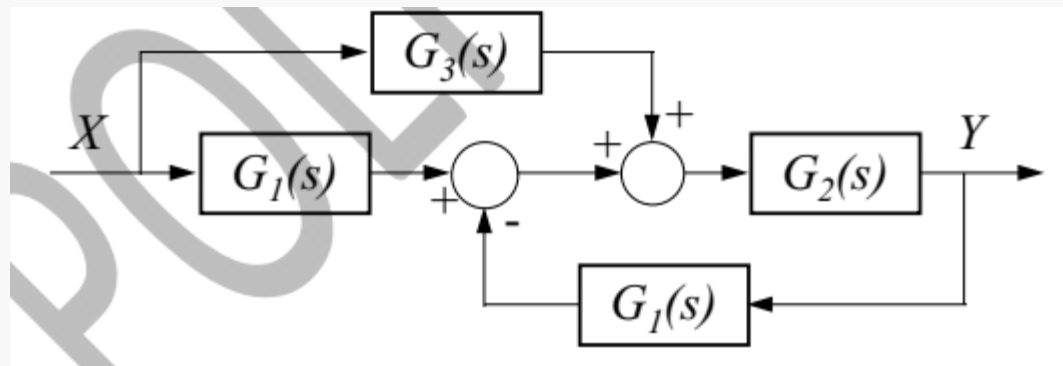
Feed-Forward



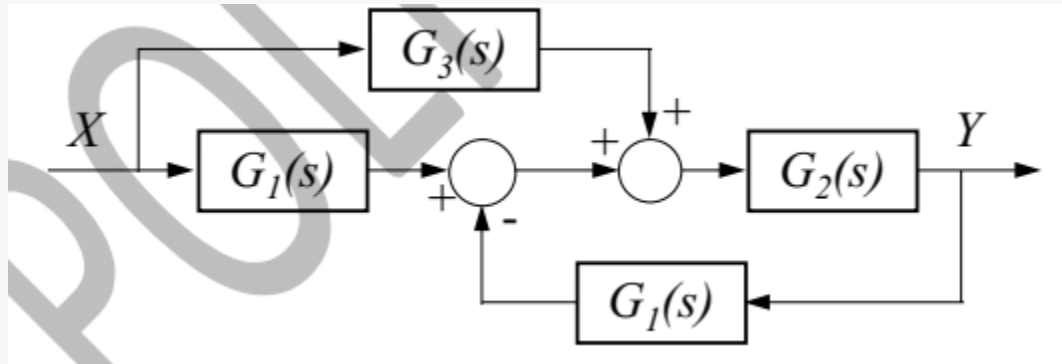
Feed-Forward



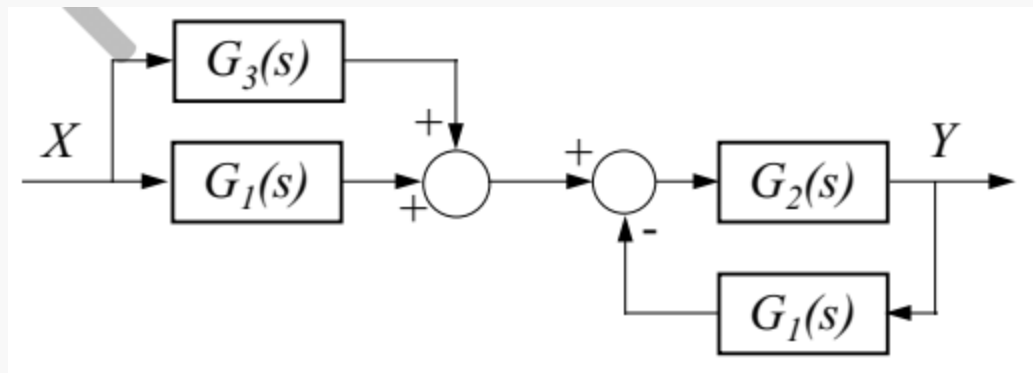
1 – Mover G_1 para antes do somador:



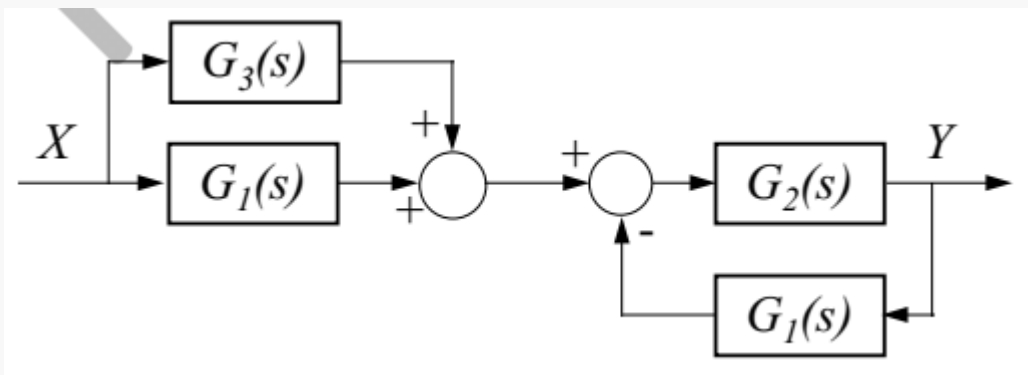
Feed-Forward



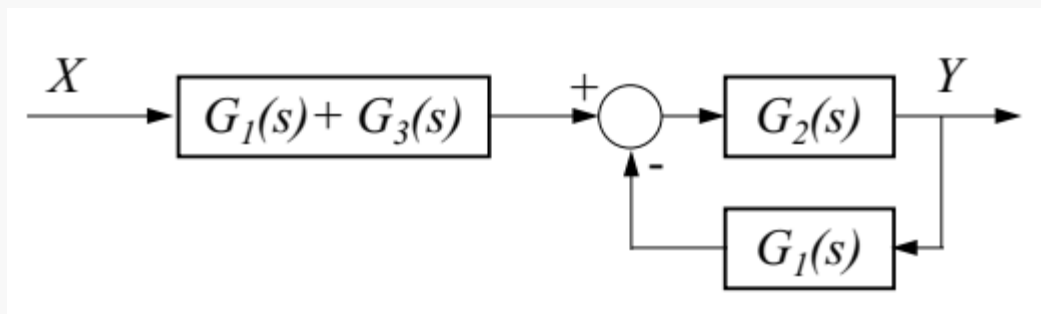
2- Rearranjar somadores:



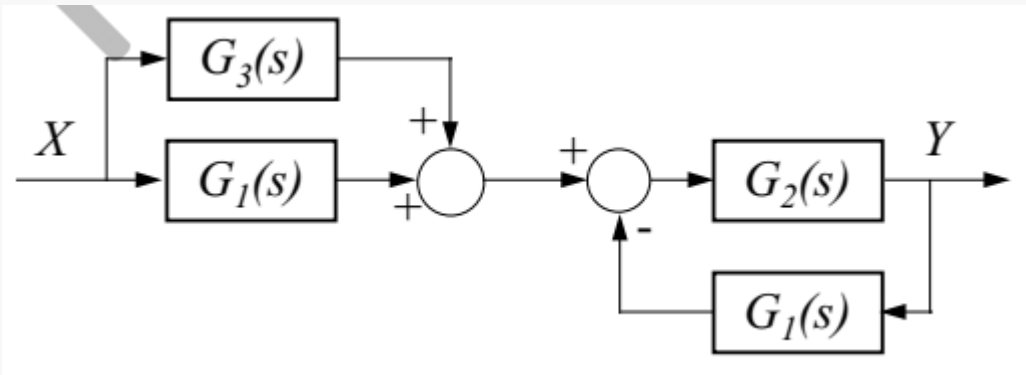
Feed-Forward



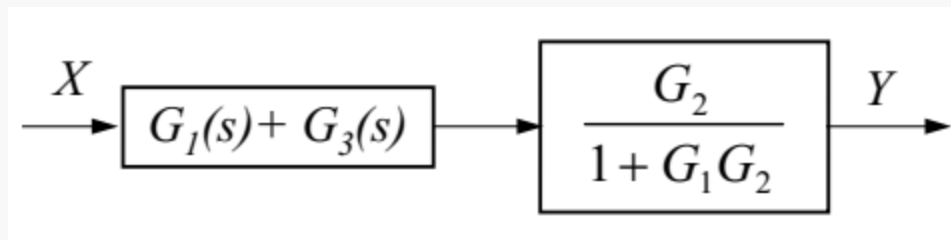
3- Efetuar o Paralelo G_3 e G_1 :



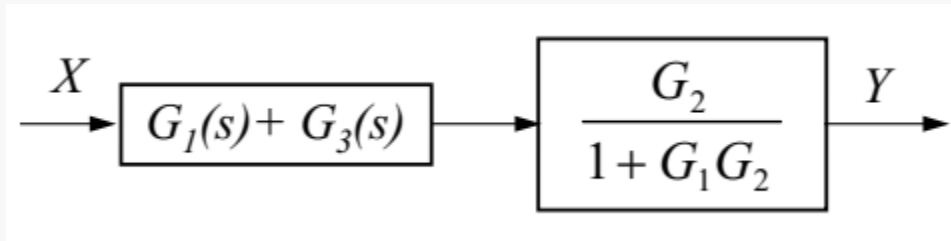
Feed-Forward



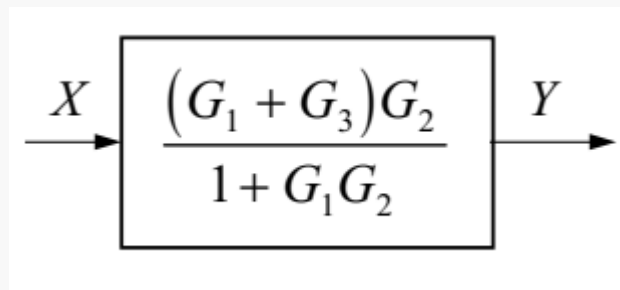
4- Efetuar o Feedback G_2 e G_1 :



Feed-Foward

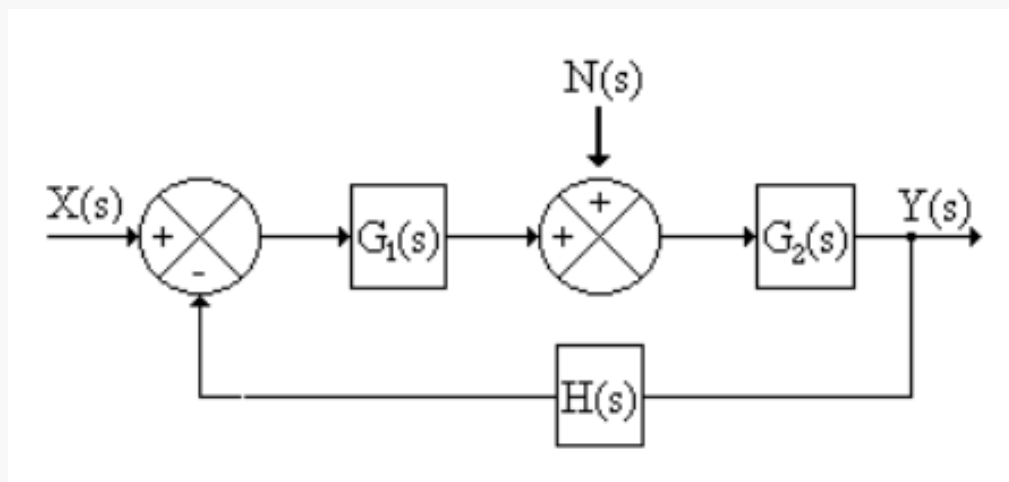


5- Efetuar a série:



$$\text{Se } G_3(s) = \frac{1}{G_2(s)}$$
$$\frac{\left(G_1(s) + \frac{1}{G_2(s)} \right) G_2(s)}{1 + G_1(s) G_2(s)} = 1$$

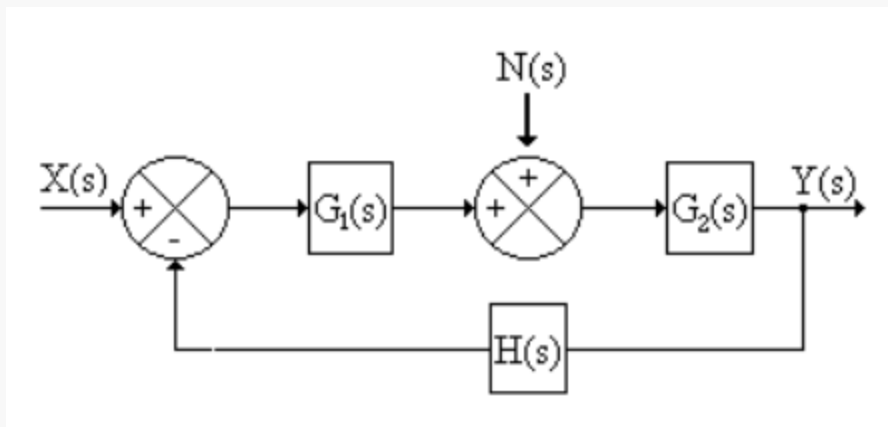
Sistemas com mais de uma entrada



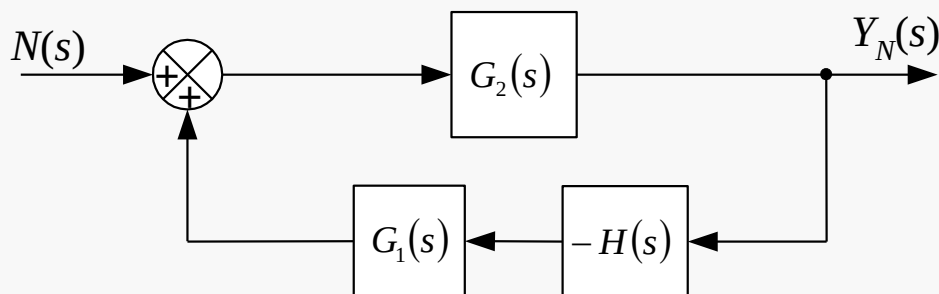
- Usa superposição:

$$Y(s) = Y_N(s) + Y_X(s)$$

Sistemas com mais de uma entrada

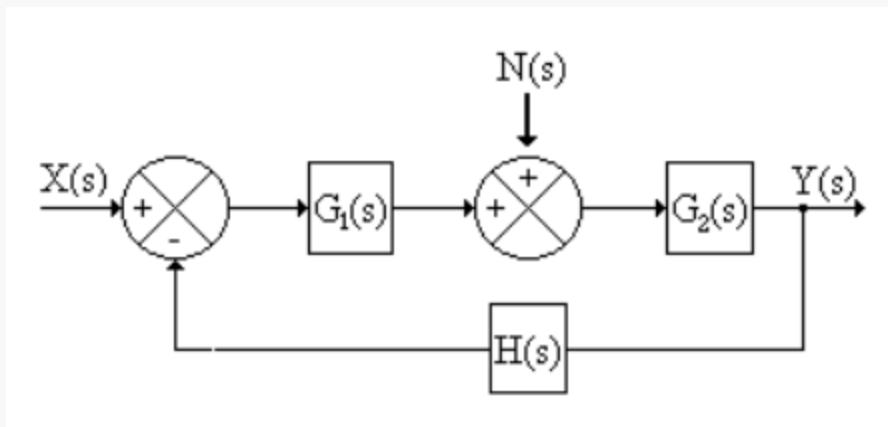


- Eliminando $X(s)$:

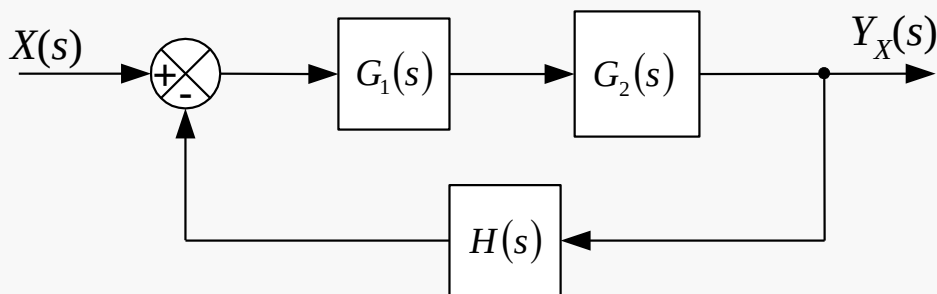


$$Y_N(s) = \frac{G_2(s)}{1 + G_2(s)G_1(s)H(s)} N(s)$$

Sistemas com mais de uma entrada

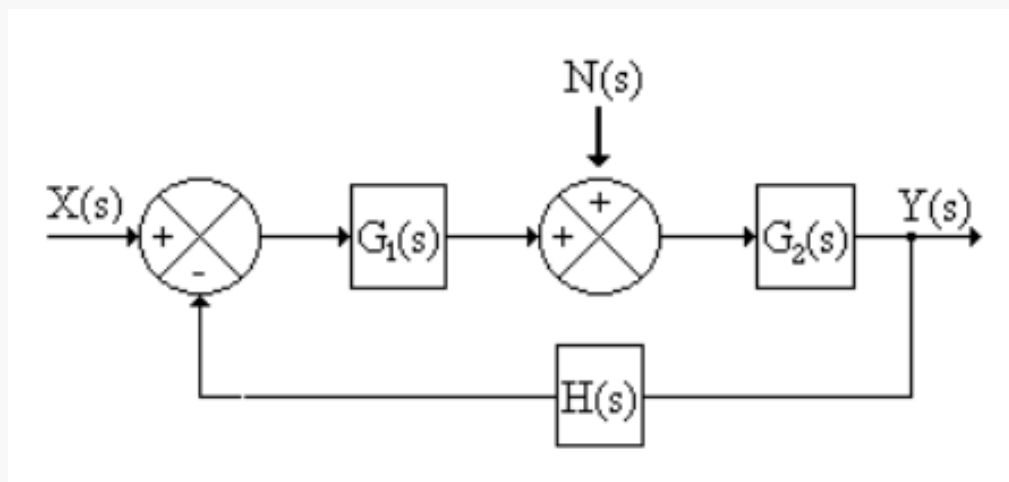


- Eliminando $N(s)$:



$$Y_N(s) = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)} X(s)$$

Sistemas com mais de uma entrada



$$Y(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} N(s) + \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)} X(s)$$

$$Y(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} [N(s) + G_1(s)X(s)]$$